Prerequisites — Part 2 of ?

Comprehensive Guide to Spherical Trigonometry, Equatorial Coordinates, Terminators, and Beta Angles

October 13, 2026

Total time spent: 8 hours and 8 minutes

This document collects the spherical trigonometry identities, equatorial coordinate conversions, terminator geometry, and beta angle relations used in spacecraft orbital mechanics. Each result is stated with the convention it depends on.

1 Spherical Trigonometry Foundations

A spherical triangle is bounded by three arcs of great circles. Its sides a, b, c are angular lengths, measured as the central angle subtended by each arc, and its angles A, B, C are the interior angles at the vertices. Unlike a plane triangle, the interior angles sum to more than 180°, and the excess is proportional to the area of the triangle.

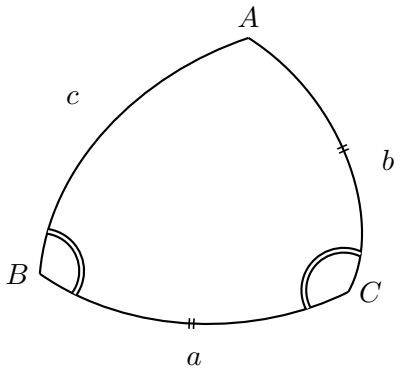


Figure 1. A spherical triangle on the unit sphere. Side a lies opposite vertex A, and so on. The sides are drawn as curves to indicate great-circle arcs.

1.1 Spherical Law of Cosines for Sides

The side a opposite angle A is determined by the other two sides and the included angle:

$$\cos a = \cos b \cos c + \sin b \sin c \cos A \tag{1}$$

As $a \to 0$ the triangle shrinks to a plane triangle, and the formula reduces to the ordinary law of cosines.

1.2 The Angle Analog

The polar triangle swaps sides and angles, which gives the dual relation for angles. The angle A is determined by the other two angles and the side a between them:

$$\cos A = -\cos B \cos C + \sin B \sin C \cos a \tag{2}$$

1.3 Spherical Law of Sines

The sines of the sides are proportional to the sines of their opposite angles:

$$\frac{\sin A}{\sin a} = \frac{\sin B}{\sin b} = \frac{\sin C}{\sin c} \tag{3}$$

1.4 Napier's Four-Part Formula

Napier's four-part formula relates two adjacent sides and two adjacent angles, taken from the four consecutive parts of the triangle. In the standard notation it reads

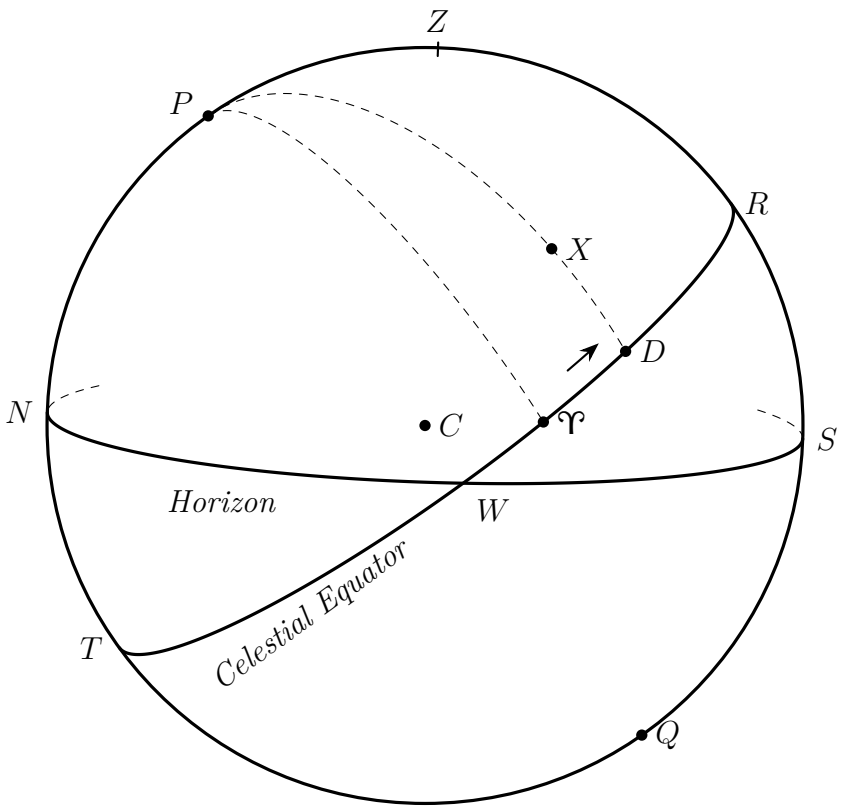
$$\cos b \cos C = \sin b \cot a - \sin C \cot A \tag{4}$$

Here b and C form the inner pair, and a and A form the outer pair. Any cyclic relabelling of the triangle gives another valid instance of the same pattern.

2 Equatorial Coordinates: Right Ascension and Declination

The equatorial system fixes positions on the celestial sphere independently of Earth's rotation. It uses the celestial equator and the vernal equinox γ as its reference points.

- **Declination δ:** the angular distance north (+) or south (−) of the celestial equator, analogous to terrestrial latitude.
- **Right ascension α:** the angular distance measured eastward along the celestial equator from the vernal equinox, analogous to terrestrial longitude. It is usually quoted in hours, with $24\ \mathrm{h} = 360^\circ$.



C: centre of sphere
X: star
D: projection of X on equator
Υ: RA zero point (vernal equinox)
P, Q: celestial north and south poles
Z: zenith
N, S: north and south points of horizon
W: intersection of horizon and equator
mutable depending on observer location
R, T: ends of the celestial equator

Each star's projection (D) on the equator serves as the 0° starting point to measure its declination

Vernal Equinox (Υ): Observer-independent, absolute cosmic point; defined by the ecliptic-equator intersection, fixed against the stellar background at epoch J2000.0 to freeze the coordinate system against annual precessional shifts (unlike the shifting March calendar event).

Figure 2. Equatorial coordinates of a star X. The declination δ is the angle from the equatorial plane up to X, and α is measured eastward along the equator from γ to the meridian through X. The dashed lines are auxiliary: F is the foot of the perpendicular from X to the equatorial plane.

$$\left[\frac{\text{JD}}{\text{Days}}\right] = \lfloor 365.25 \times (Y + 4716) \rfloor + \lfloor 30.6001 \times (M + 1) \rfloor + \left[\frac{D}{\text{Days}}\right] + B - 1524.5 + \left[\frac{\text{UT}}{\text{Days}}\right] \quad (5)$$

Where:

if $M \leq 2$, then $Y = Y - 1$ and $M = M + 12$.

$$A = \lfloor Y/100 \rfloor$$

$$B = 2 - A + \lfloor A/4 \rfloor$$

$$\left[\frac{\text{UT}_{\text{hours}}}{\text{Hours}}\right] = \text{Local Time} - \text{Time Zone Offset}$$

$$\left[\frac{\mathrm{UT_{hours}}}{\mathrm{Hours}}\right] = \mathrm{Hours} + \frac{\mathrm{Minutes}}{60} + \frac{\mathrm{Seconds}}{3600}$$

Evaluation for J2000.0 (January 1, 2000, 12:00 UT):

Given Inputs: $Y = 2000$, $M = 1$, $\left[\frac{D}{\mathrm{Days}}\right] = 1$, $\left[\frac{\mathrm{UT_{hours}}}{\mathrm{Hours}}\right] = 12^{\mathrm{h}} \implies \left[\frac{\mathrm{UT_{days}}}{\mathrm{Days}}\right] = \frac{12}{24} = 0.5\,\mathrm{days}$.

1. Adjust Month and Year ($M \leq 2$):

$$Y = 2000 - 1 = 1999, \quad M = 1 + 12 = 13$$

2. Calculate Gregorian Correction Factors (A and B):

$$A = \lfloor 1999/100 \rfloor = 19$$

$$B = 2 - 19 + \lfloor 19/4 \rfloor = 2 - 19 + 4 = -13$$

3. Substitute into the JD Equation:

$$\left[\frac{\mathrm{JD}}{\mathrm{Days}}\right] = \lfloor 365.25 \times (1999 + 4716) \rfloor + \lfloor 30.6001 \times (13 + 1) \rfloor + 1 + (-13) - 1524.5 + 0.5$$

$$\left[\frac{\mathrm{JD}}{\mathrm{Days}}\right] = \lfloor 365.25 \times 6715 \rfloor + \lfloor 30.6001 \times 14 \rfloor - 12 - 1524.5 + 0.5$$

$$\left[\frac{\mathrm{JD}}{\mathrm{Days}}\right] = 2451233 + 428 - 1536.5 + 0.5 = \mathbf{2451545.0}$$

2.1 Conversion to Local Altitude

To convert equatorial coordinates to the local horizon system, let ϕ be the observer's latitude, δ the object's declination, and H the local hour angle. The altitude h of the object satisfies the fundamental relation:

$$\sin([h/\mathrm{deg}]) = \sin\phi \sin\delta + \cos\phi \cos\delta \cos\left([H/\mathrm{hours}] \times 15^\circ\right) \quad [\mathrm{Dimensionless}] \tag{6}$$

Section 2.1.2

2.1.1 The Local Hour Angle (H)

The Local Hour Angle (H) describes how far west (or east) an object is positioned relative to the local meridian—the great circle passing through the observer's zenith from north to south. It represents the angular distance measured westward along the celestial equator from the observer's meridian to the hour circle of the object, expressed in decimal hours and defined by:

$$\left[\frac{H}{\mathrm{Hours}}\right] = \left[\left[\frac{\mathrm{LST}}{\mathrm{Hours}}\right] - \left[\frac{\alpha}{\mathrm{Hours}}\right]\right] \pmod{24}$$

where LST is the Local Sidereal Time and α is the Right Ascension of the celestial object, both in decimal hours.

Step-by-Step Guide to Finding Each Component (All in Decimal Hours):

1. **Find the Right Ascension (α) of the Object:** Obtain the coordinate from a standard star catalog or ephemeris. Right Ascension must be converted from hours, minutes, and seconds into **decimal hours** for mathematical consistency:

$$\left[\frac{\alpha}{\text{Hours}}\right] = \text{Hours} + \frac{\text{Minutes}}{60} + \frac{\text{Seconds}}{3600}$$

2. **Find the Local Sidereal Time (LST):** Sidereal time measures Earth's rotation relative to the stellar background. All inputs and outputs are strictly in decimal hours. The Local Sidereal Time is computed through sequential corrections:

   - **Convert Local Time to Universal Time (UT):**

$$\left[\frac{\text{UT}}{\text{Hours}}\right] = \text{Local Time} - \text{Time Zone Offset}$$

   - **Determine Greenwich Sidereal Time (GST):** Calculate Greenwich Mean Sidereal Time directly from the days elapsed since the J2000.0 epoch ($D = \text{JD} - 2451545.0$), expressed in decimal hours and normalized using a 24-hour cycle:

$$\left[\frac{\text{GST}}{\text{Hours}}\right] = \left[6.697374558 + 0.06570982441908 \times \left[\frac{D}{\text{Days}}\right] + 1.00273790935 \times \left[\frac{\text{UT}}{\text{Hours}}\right]\right] \pmod{24}$$

   - **Apply Longitude Correction (λ):** Convert geographic longitude λ into time units ($15^\circ = 1$ hour), taking East as positive and West as negative, and add it to GST with a 24-hour wrap:

$$\left[\frac{\text{LST}}{\text{Hours}}\right] = \left[\left[\frac{\text{GST}}{\text{Hours}}\right] + \left[\frac{\lambda}{15^\circ/\text{h}}\right]\right] \pmod{24}$$

3. **Compute the Greenwich Hour Angle (GHA):** Subtract the decimal Right Ascension (α) from the Greenwich Sidereal Time, expressed in decimal hours and normalized to the 0^h to 24^h range:

$$\left[\frac{\text{GHA}}{\text{Hours}}\right] = \left[\left[\frac{\text{GST}}{\text{Hours}}\right] - \left[\frac{\alpha}{\text{Hours}}\right]\right] \pmod{24}$$

4. **Compute the Local Hour Angle (H):** Subtract the decimal Right Ascension from the Local Sidereal Time, expressed in decimal hours and normalized:

$$\left[\frac{H}{\text{Hours}}\right] = \left[\left[\frac{\text{LST}}{\text{Hours}}\right] - \left[\frac{\alpha}{\text{Hours}}\right]\right] \pmod{24}$$

   - **Signed Format Rule:** To convert into the traditional $\pm 12^\text{h}$ range (where negative values represent positions east of the meridian and positive values represent positions west), subtract 24^h if the normalized hour angle exceeds 12^h.

Interpretation of H:

- $H = 0$h: The object lies directly on the local meridian (transit / culmination).
- $H > 0$h (e.g., +3h): The object has crossed the meridian and is west of it.
- $H < 0$h or near 24h (e.g., 21h / −3h): The object is east of the meridian, approaching transit.

2.1.2 Derivation via the Spherical Law of Cosines

This formula is derived by applying the spherical law of cosines to the astronomical triangle (the PZX triangle) formed by three key reference points on the celestial sphere:

- P **(North Celestial Pole):** The pivot point around which the sky appears to rotate.
- Z **(Zenith):** The point located directly overhead the observer.
- X **(Celestial Object):** The star or astronomical body being tracked.

P is the Pivot point where Hour angle is measured from

The general spherical law of cosines for a side c enclosed by sides a and b and the included angle C states:

$$\cos c = \cos a \cos b + \sin a \sin b \cos C$$

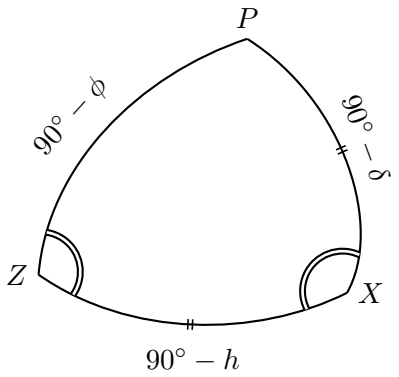
When substituting the components of our astronomical triangle (PZX):

- The side opposite to angle H is $c = ZX = 90^\circ - h$ (Zenith distance / co-altitude)
- The first side is $a = PZ = 90^\circ - \phi$ (Co-latitude)
- The second side is $b = PX = 90^\circ - \delta$ (Co-declination)
- The included angle is $C = H$ (Local Hour Angle)

Substituting these values into the law of cosines yields:

$$\cos(90^\circ - h) = \cos(90^\circ - \phi)\cos(90^\circ - \delta) + \sin(90^\circ - \phi)\sin(90^\circ - \delta)\cos H$$

Using trigonometric co-function identities ($\cos(90^\circ - \theta) = \sin\theta$ and $\sin(90^\circ - \theta) = \cos\theta$), this simplifies directly to the final altitude formula.



Local Hour Angle (LHA): The angular distance of a celestial object measured westward along the celestial equator from the observer's local meridian. The local meridian is defined as the great circle passing through the observer's Zenith, Nadir, and both the North and South Celestial Poles.

Greenwich Hour Angle (GHA / HA): The same angular distance measured westward along the celestial equator from the Prime Meridian. The Prime Meridian is the reference great circle passing through Greenwich, England (0° longitude) and both the North and South Celestial Poles.

3 The Terminator

The terminator is the boundary between the illuminated day side and the dark night side of a planetary body. It is the set of points where the solar zenith angle θ_z equals 90°, so $\cos\theta_z = 0$.

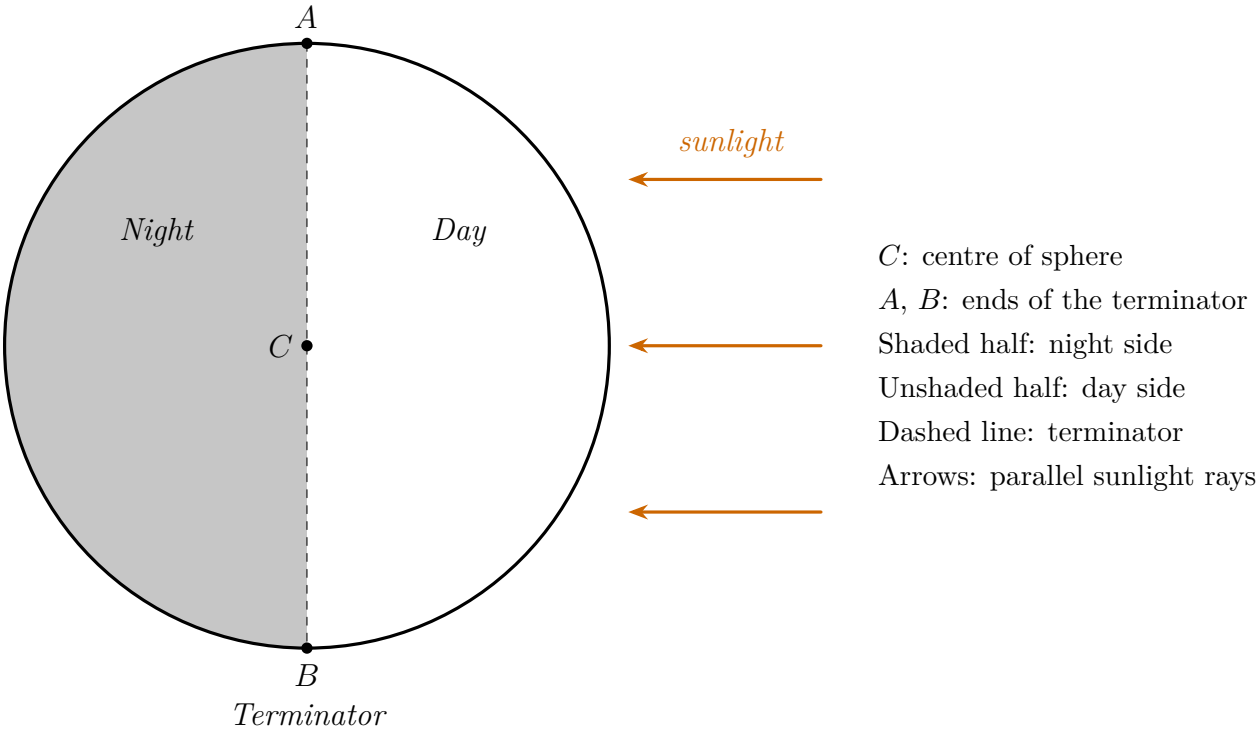


Figure 3. The terminator for a body whose subsolar point lies on the equator. Here the terminator is a great circle through the poles, which appears as the dashed vertical line.

3.1 Terminator Equation

For a point at latitude ϕ with local solar hour angle $[h_s/\text{hours}]$, when the Sun has declination $\delta_\odot$,

$$\cos\left([\theta_z/\text{deg}]\right) = \sin\phi\sin\delta_\odot + \cos\phi\cos\delta_\odot\cos\left([h_s/\text{hours}]\times 15^\circ\right) \quad \text{[Dimensionless]} \tag{7}$$

Setting $\cos\theta_z = 0$ gives the terminator.

3.2 Latitude–Hour Angle Relation

Solving $\cos\theta_z = 0$ for the latitude gives

$$\tan\left([\phi/\text{deg}]\right) = -\frac{\cos\left([h_s/\text{hours}]\times 15^\circ\right)}{\tan\left([\delta_\odot/\text{deg}]\right)} \quad \text{[Dimensionless]} \tag{8}$$

When $\delta_\odot = 0^\circ$, the right-hand side diverges unless $\cos\left([h_s/\text{hours}]\times 15^\circ\right) = 0$. The terminator then consists of the meridians at $[h_s/\text{hours}] = \pm 6^\text{h}$ (corresponding to $\pm 90^\circ$), which matches Figure 3.

3.3 Solar Declination Computation Methods

The solar declination $\delta_\odot$ can be determined through two primary approaches:

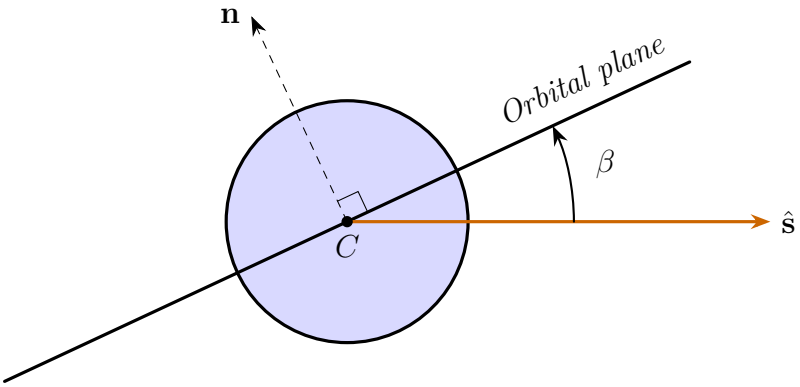
1. **The Approximate Method:** Uses Cooper's equation based on the sequential day of the year n for standard estimations:

$$\delta_\odot = 23.45^\circ \sin\left(\frac{360^\circ}{365}\left([n/\text{days}] + 284\right)\right) \quad \text{[Degrees]}$$

2. **The Advanced Method:** Utilizes rigorous ecliptic coordinate transformations accounting for Earth's orbital eccentricity and obliquity: *I will study it later once I've time!*

4 Beta Angle and Eclipse Conditions

In astrodynamics, the beta angle β is the angle between a spacecraft's orbital plane and the direction to the Sun. It controls the thermal environment of the spacecraft and the length of its eclipses.



C: centre of sphere
$\hat{\mathbf{s}}$: unit vector toward the Sun
$\mathbf{n}$: normal to the orbital plane
β: angle between $\hat{\mathbf{s}}$ and the orbital plane

Figure 4. Edge-on view of the orbital plane. The vector $\mathbf{n}$ is the orbit normal, and β is the angle between the plane and the Sun direction $\hat{\mathbf{s}}$.

4.1 Beta Angle in Terms of Orbital Elements

Write β through the dot product of the unit orbit normal $\mathbf{n}$ and the unit Sun vector $\hat{\mathbf{s}}$:

$$\sin\beta = \mathbf{n}\cdot\hat{\mathbf{s}} \quad \text{[Dimensionless]}$$

Expressed in terms of its Cartesian components within the equatorial frame, the orbit normal is determined by the inclination i and the right ascension of the ascending node Ω:

$$\mathbf{n} = (\sin i \sin\Omega,\ -\sin i\cos\Omega,\ \cos i)$$

The Sun direction has components $\hat{\mathbf{s}} = (\cos\delta_\odot\cos\alpha_\odot,\ \cos\delta_\odot\sin\alpha_\odot,\ \sin\delta_\odot)$. Writing out the dot product explicitly component-by-component gives:

$$\sin\beta = (\sin i\sin\Omega)(\cos\delta_\odot\cos\alpha_\odot) + (-\sin i\cos\Omega)(\cos\delta_\odot\sin\alpha_\odot) + (\cos i)(\sin\delta_\odot) \quad \text{[Dimensionless]}$$

To prove and simplify this expression, factor out $\sin i\cos\delta_\odot$:

$$\sin\beta = \sin i\cos\delta_\odot\Big(\sin\Omega\cos\alpha_\odot - \cos\Omega\sin\alpha_\odot\Big) + \cos i\sin\delta_\odot \quad \text{[Dimensionless]}$$

Using the trigonometric angle subtraction identity $\sin(\Omega-\alpha_\odot) = \sin\Omega\cos\alpha_\odot - \cos\Omega\sin\alpha_\odot$ and applying the property $\sin(\Omega-\alpha_\odot) = -\sin(\alpha_\odot-\Omega)$, we arrive at the final form:

$$\sin\beta = \sin\delta_\odot\cos i - \cos\delta_\odot\sin i\,\sin(\alpha_\odot-\Omega) \quad \text{[Dimensionless]} \tag{9}$$

Reversing the sign convention for $\mathbf{n}$ flips the sign of β. Eclipse conditions depend only on $|\beta|$, so the magnitude is what matters.

4.2 Critical Beta Angle and Eclipses

A satellite in a circular orbit of altitude h around a body of radius R_E just grazes the shadow when

$$\beta^* = \arcsin\left(\frac{R_E}{R_E+h}\right) \quad \text{[Degrees]} \tag{10}$$

The eclipse condition follows directly from comparing the two states: you use the first equation (9) to find out where your satellite is (β), and the second equation (10) to find out where the shadow boundary is (β^*). Comparing them reveals whether you dip into the shadow.

- $|\beta| < \beta^*$**:** the orbit intersects the planetary shadow, so the spacecraft is eclipsed once per orbit.
- $|\beta| \geq \beta^*$**:** the orbit stays entirely in sunlight, and the eclipse duration is zero.